

medical emergencies, pandemics, or epidemic scenarios. First, both DCNNs and RNNs take less than a few seconds to diagnose medical conditions and monitor disease progression. The accelerated speed of diagnosis makes way for early treatment intervention and better cure rates. Second, it helps radiology experts optimise their time better and effectively work alongside deep learning networks. They can automate repetitive and time-consuming imaging reading tasks to AI-based neural networks and focus more on developing their medical domain expertise.

IoT integrated primary care

Primary healthcare means monitoring patients' responses to medical treatments during doctor visits or hospital stays. However, with IoT, the frequency of all such in-person visits can be significantly reduced. Even better, all the health stats can be accurately monitored remotely. Diagnostic wearable sensors like fitness bands, pulse oximeters, fluid level meters, or sphygmomanometers; nutrition monitoring, and sleep detectors can be used to collect critical patient data. All analogue data collected from sensors can be aggregated and converted into digital formats by the analogue-to-digital converters (ADCs) integrated into the IoT devices. The aggregated digital data is further pre-processed and published to data servers or the service cloud using HTTP or MQTT protocols. All repository data from the cloud can be used to create analytics instances using RESTful API services. Developers can perform real-time health telemetry streaming, detect pathological events, gain insights into historical patient records, and give information about diseases affecting multiple organs. The application of ML algorithms on health data sets can reveal hidden pathological correlations, raise alarms in case of

abnormalities, and provide predictive health analytics. Post-visit medication adherence, treatment progress, and disease cure can all be tele-monitored using IoT devices and wearable biosensors.

Centralised disease tracking applications

As of today, health information systems are largely limited to hospital administration. To some extent, insurance companies do keep track of the disease history of their policyholders. However, a large-scale centralised application to maintain citizens' healthcare assets can facilitate better public healthcare services at a macro-scale. IoT and RFID can be integrated to create and maintain all births, vaccinations, diseases, and death records. Centralised healthcare applications can help doctors access all disease records of their patients on a single user interface without them having to review papers or verbal declarations manually. Electronic Health Records (EHRs) can benefit doctors by providing a better prognosis and treatment via tele-health. The only thing that is challenging the development of such applications today is privacy and cybersecurity. Automation is indeed a double-edged sword. While there are many benefits of automating health records, the risk of hacking, data breach, and ransomware does exist. Digitisation of medical records requires high-level application security, compliance audits, and security automation. It also requires stringent laws and regulations for personal data protection — something along the lines of the European General Data Protection Regulation (GDPR). If technology professionals and those in governance can address this risk, then a centralised server can be the ideal solution to control all disease outbreaks, and not just this pandemic.

In the case of COVID-19, fatalities were more evident for those with some disease history like diabetes, blood pressure, obesity, myocardial infarctions, or renal problems. An AI-IoT integrated application could quickly identify these highly vulnerable groups so that customised critical care could be offered to these patients. COVID fatalities could have been greatly minimised had there been an AI-IoT based approach to primary healthcare. Many in the technology industry have resisted AI this far in the fear of job loss. However, AI will not complement jobs; it will supplement them. With the help of AI, professionals in information technology, medical, or those in governance can focus on their core expertise while making automation and AI do all the repetitive and analytical work for them.

Across the globe, we claim to be a technologically advanced and a digitally competent generation. However, are we able to better control disasters or disease outbreaks? The global COVID-19 statistics prove that we can't, yet.

This is a wake-up call, and the need of the hour is for us to take advantage of all the technology that is available to us. Artificial intelligence, machine learning, the Internet of Things (IoT), and automation can help medical practitioners provide better patient care. They can help public policymakers build socially resilient communities. Technology and digital transformation can help us be better prepared for the next pandemic and thereafter. So, in the best interest of humanity, let us adopt AI power for public health. **END** 🐧

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